

BOKARO STEEL PLANT — ILLUSTRATIVE SAMPLE

Standard Operating Procedure

SOP No: BSP-SMS-012

Title: Oxygen Blowing Operation in LD Converter

Department: Steel Melting Shop

Revision: 02

Date: 05-Jun-2024

1. PURPOSE

To define the safe procedure for oxygen blowing in the LD (Linz-Donawitz) converters of Steel Melting Shop-II.

2. SCOPE

Applicable to LD Converters A, B and C. To be followed by converter operators, blowing pulpit operators and shift in-charges.

3. PREREQUISITES

- Lance cooling water flow minimum 180 m³/h and outlet temperature below 50°C
- Oxygen header pressure between 14 and 16 kg/cm²
- LD gas recovery system (IDCS) in auto mode and gas holder level below 80%
- Converter mouth and hood area cleared of all personnel
- Scrap and hot metal charged as per heat chart; slag-forming fluxes ready in bunkers

4. PERSONAL PROTECTIVE EQUIPMENT (PPE)

- Aluminised heat-resistant jacket and trousers
- Face shield with IR filter (shade 5)
- Safety helmet with neck protector
- Portable CO monitor (personal) for anyone on the converter platform

5. BLOWING PROCEDURE

Step 1 — Lower the lance to 2.5 m above bath level and start oxygen at 50% flow for ignition.

Step 2 — Confirm ignition within 60 seconds from the flame at the converter mouth. If there is no ignition within 60 seconds, raise the lance and stop oxygen.

Step 3 — Increase oxygen flow to 100% (approx. 600 Nm³/min) and bring the lance to 1.8 m working height.

Step 4 — Add lime and dolomite in two batches as per the flux chart during the first 4 minutes.

Step 5 — Watch for slopping. On heavy slopping, raise the lance by 0.3 m and reduce oxygen flow to 80%.

Step 6 — Stop blowing at the end point given by sub-lance or after approx. 16 minutes; raise the lance to parking position.

6. EMERGENCY — LANCE WATER LEAK

- Stop oxygen immediately and raise the lance to parking position
- Do NOT tilt the converter until water on the bath surface has fully evaporated (minimum 20 minutes)
- Water contact with liquid steel can cause a steam explosion
- Inform Shift In-charge (Ext. 3120) and Emergency Control Room (Ext. 100)

7. RECORDS

Record blow time, oxygen consumed and any abnormality in the Heat Log Book (form SMS-HL-02).